Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 12 Tutorial Worksheet 1: Cryptography Fundamentals

Field	Detail
Based on	Week 12 Deck A, Cryptography Fundamentals lecture (foundations, classical ciphers, symmetric encryption, block vs stream ciphers, asymmetric encryption, hybrid encryption)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate and compare security technologies)
Format	15 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's Cryptography Fundamentals lecture. Several ask you to calculate, recommend, or evaluate rather than simply describe, which is deliberate: LO3 and LO4 are about choosing the right technology for a requirement, not just naming it. Write full answers, not one-line definitions.

Foundations

Q1. State Kerckhoffs's principle, and explain why a cipher whose design is public but whose key is secret is considered stronger than a cipher whose design is kept secret. *[LO4]*

Q2. Symmetric encryption protects confidentiality. Explain which other security services (integrity, authentication, non-repudiation) it does not provide on its own, and why. *[LO3]*

Classical Ciphers

Q3. Encrypt the word NETWORK using a Caesar cipher with a shift of 5. Show your working for each letter, and then explain how an attacker who only sees the ciphertext could recover the message without knowing the shift. *[LO3]*

Q4. Explain why the Vigenere cipher defeats simple frequency analysis, yet is still breakable. What do the weaknesses of classical ciphers teach us about the properties a modern cipher must have? *[LO4]*

Symmetric Encryption

Q5. A company has 200 employees, and every pair of employees needs a private symmetric channel. Calculate how many keys are needed using n(n-1)/2, and explain the two separate problems that make symmetric key distribution hard. *[LO3]*

Block vs Stream Ciphers

Q6. Explain what a mode of operation is, and explain why ECB mode is unsuitable for encrypting a long message while CBC mode is better. *[LO4]*

Q7. Explain how CTR mode makes a block cipher behave like a stream cipher, and state the single most important rule for using it safely. *[LO4]*

Q8. Using WEP as your example, explain why reusing a keystream is catastrophic. Include what an attacker learns when two messages are encrypted with the same keystream. *[LO3]*

Q9. Compare block ciphers and stream ciphers on four aspects: the unit of data processed, the need for padding, what happens when a transmission error occurs, and the characteristic way each tends to fail in practice. *[LO4]*

Q10. Recommend a cipher family and mode for each of the following, and justify each choice: (a) full disk encryption on a laptop, (b) a real-time voice call, (c) a TLS session on a low-end phone with no AES hardware support. *[LO3, LO4]*

Asymmetric Encryption

Q11. Explain, in your own words, how Diffie-Hellman lets two parties who have never met agree a shared secret over a public channel, and explain why it is vulnerable to a man in the middle when used on its own. *[LO3, LO4]*

Q12. A company is designing a battery-powered sensor with very little memory and processing power that must sign its readings. Compare RSA and elliptic curve cryptography for this device and recommend one, justifying your answer. *[LO3, LO4]*

Q13. Give two advantages and two disadvantages of asymmetric encryption compared with symmetric encryption. *[LO4]*

Hybrid Encryption

Q14. Explain why neither symmetric nor asymmetric encryption is sufficient on its own, and describe the two phases of hybrid encryption using a TLS connection as your example. *[LO3, LO4]*

Q15. A colleague proposes encrypting all traffic in a new VPN directly with RSA, to avoid having to manage session keys. Evaluate this proposal and explain what would go wrong. *[LO4]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. Several questions ask you to recommend one option for a specific scenario. Practise stating your recommendation first, then justifying it against the scenario's actual requirements, since that is exactly the structure a structured-essay exam question rewards.
3. Carry one open question forward into the next worksheet: when you receive a public key, how do you know it belongs to who it claims to? Today's questions on Diffie-Hellman and hybrid encryption are incomplete without the answer.
4. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5).